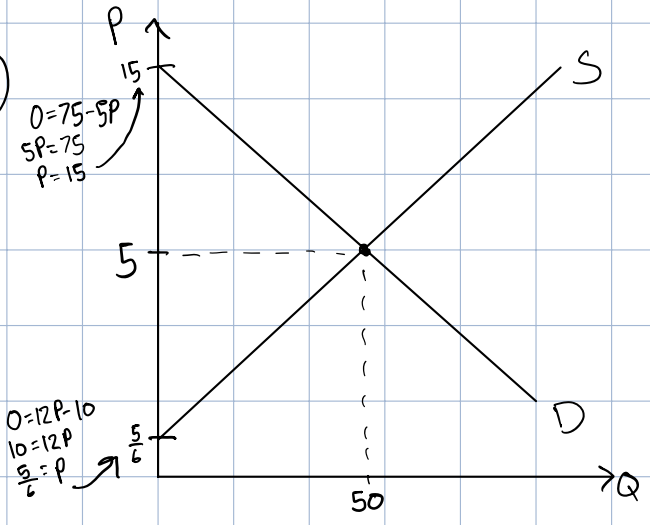
Solns

1.a)



$75-5P^* = 12P^*-10$

$85 = 17P^*$

$5 = P^*$

$Q^* = 75-5(5)$

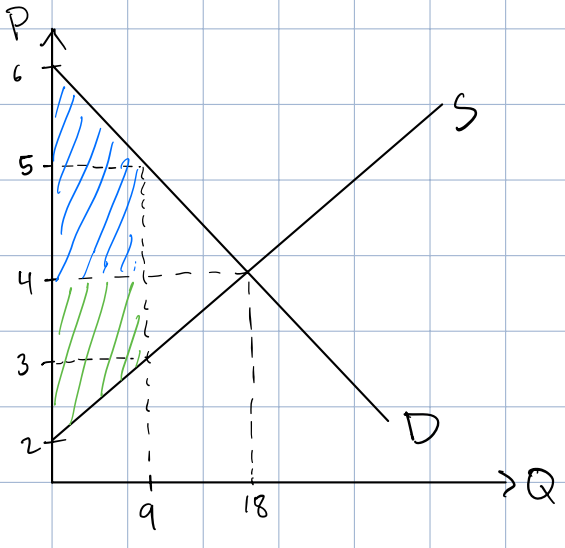
$= 50$

b) $TS = \frac{1}{2}(15-\frac{5}{6})(50) = \frac{85}{6}(25) = \frac{2,125}{6} \approx 354$

2.a) $CS = \frac{1}{2}(6-4)(18m) = 18$ million

$PS = \frac{1}{2}(4-2)(18m) = 18$ million

b.

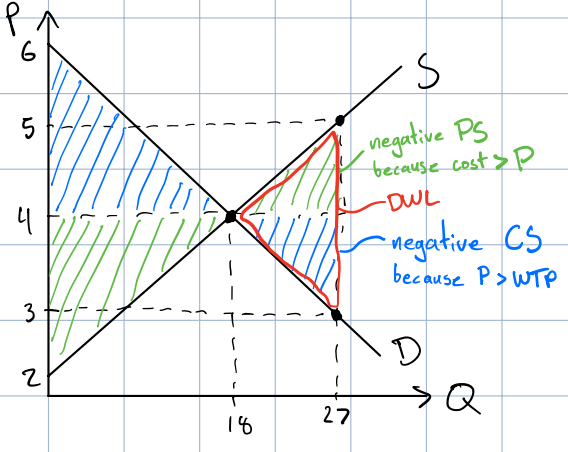


$CS = \frac{1}{2}(6-5)(9m) + (5-4)(9m) = 13.5$ million

$PS = \frac{1}{2}(3-2)(9m) + (4-3)(9m) = 13.5$ million

$DWL = \frac{1}{2}(5-3)(18m-9m) = 9$ million

c.

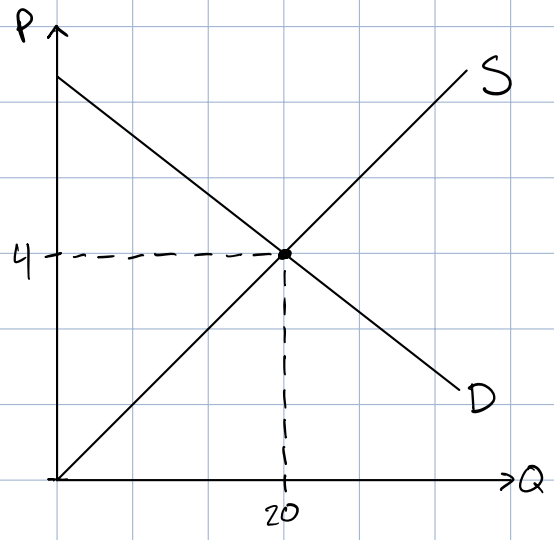


$CS = \frac{1}{2}(6-4)(18m) - \frac{1}{2}(4-3)(27m-18m) = 13.5$ mil

$PS = \frac{1}{2}(4-2)(18m) - \frac{1}{2}(5-4)(27m-18m) = 13.5$ mil

$DWL = \frac{1}{2}(5-3)(27m-18m) = 9$ mil

3.a.



$60 - 10P^* = 5P^*$

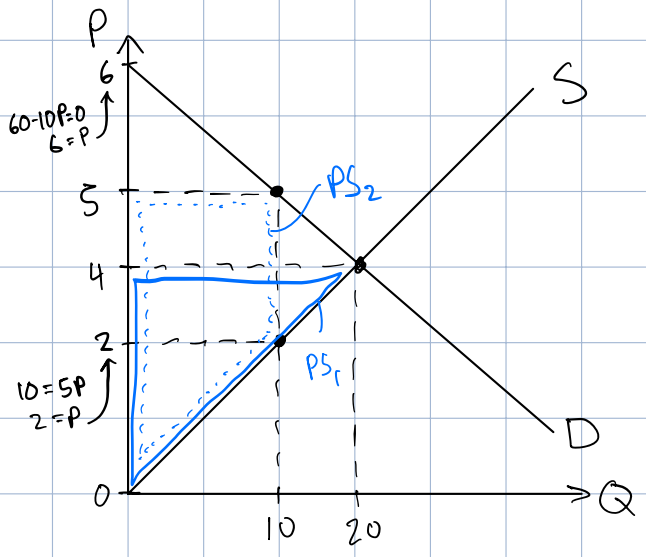
$60 = 15P^*$

$4 = P^*$

$Q^* = 5(4) = 20$

A price floor of \$4 is non-binding, so PS doesn't change.

b.



$Q_{floor} = 60 - 10(5)$

$= 10$

$PS_1 = \frac{1}{2}(4-0)(20-0) = 40$

$PS_2 = \frac{1}{2}(2-0)(10) + (5-2)(10) = 10 + 30 = 40$

Producer surplus stays the same, but who accrues it changes. The producers who can still find buyers accrue more than they used to, & the producers who used to have buyers but now don't lose all of their surplus.

4. a) $PS = \frac{1}{2}(2-0.35)(1.65b) = 1.36125$ billion

$CS = \frac{1}{2}(3.65-2)(1.65b) = 1.36125$ billion

$TS = CS + PS = 2.7225$ billion

b) $CS = \frac{1}{2}(3.65-2.05)(1.6b) = 1.28$ billion

c) $PS = \frac{1}{2}(2.05-0.35)(1.7b) = 1.445$ billion

d) $2.05(1.70b - 1.60b) = 0.205$ billion

e) TS = CS + PS - Govt Spending

= 1.28b + 1.445b - 0.205b

= 2.52 billion